

YUVALTERN

VIRTUAL CONSTRUCTION BUSINESS INTELLIGENCE INTERNSHIP

TASK 01

STRATEGY AND PLANNING

Power BI Strategy for NYC Capital Projects

Business Intelligence & Performance Analysis

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PROJECT FOCUS

Construction Project Data Analysis & Business Intelligence

Tool: Microsoft Power BI | **Data Source:** NYC Open Data – Capital Projects

Strategic Planning & Project Foundation

1. Executive Summary

The purpose of this task is to prepare a practical Business Intelligence strategy for a construction project environment. The strategy focuses on using Microsoft Power BI to bring project budget, spending and schedule information into one clear view.

For this project, I selected the NYC Capital Projects Dashboard public datasets from NYC Open Data as the main data foundation. The datasets contain information related to capital project budgets, spending, project phases and schedules. They also include historical budget/spend and schedule information, which will be useful for the analysis.

2. Task Objective

The objective of this task is to establish a practical Business Intelligence strategy for a virtual construction environment. The plan considers construction industry needs, suitable data sources, business KPIs, Power BI use cases, data preparation and reporting requirements.

1. Understand the type of information required for construction project monitoring.
2. Identify a realistic public construction data source for the project.
3. Define business objectives and useful construction KPIs.
4. Plan how Power BI can be used for reporting and decision support.
5. Identify practical construction BI use cases.
6. Keep the project realistic by separating planned analysis from actual results.

3. Project Scope

Area	Included	Not Assumed
Financial performance	Budget, spending, budget utilization and historical budget changes	Actual savings or ROI
Schedule performance	Project phase, completion dates and schedule changes	Unverified delay causes
Portfolio analysis	Agency, borough, project category and project-level comparison	Internal company rankings
Predictive analysis	Future risk analysis if the historical data supports it	Guaranteed prediction accuracy
Workforce / safety / procurement	Possible future extensions	Invented employee, safety or supplier data

4. Data Foundation – NYC Capital Projects

The main source selected for this project is the NYC Capital Projects Dashboard available through NYC Open Data. The dataset covers major capital infrastructure projects with committed budgets and combines financial and agency schedule information using the FMS ID.

The current Citywide Budget and Schedule dataset includes fields such as reporting period, managing agency, sponsor agency, FMS ID, total budget, spend to date, project name, current phase, forecast completion and borough. NYC Open Data also provides historical budget/spend and schedule datasets that can be used in time-based analysis.

5. Market and Construction BI Context

Construction projects generate information from different activities such as finance, project scheduling, procurement, site work and project administration. A manager normally needs a simple view of whether a project is within its expected budget and whether its schedule is moving as planned.

Business Intelligence can help by bringing this information into a common reporting environment. The focus of this project is not to replace the source systems. Instead, Power BI will be used as a reporting and analysis layer that makes the information easier to understand.

The selected NYC data is useful for this internship because it is public, realistic and directly related to capital projects. It also provides historical information, allowing the project to move from basic reporting toward trend and risk analysis.

6. Business Objectives

No.	Objective	How it will be used
1	Improve project cost visibility	Provide a clear view of project budget, spending, remaining budget and budget changes.
2	Improve schedule monitoring	Track project phase, forecast completion, actual completion where available and schedule changes.
3	Identify projects needing attention	Use simple indicators to highlight unusual budget or schedule movements.
4	Support comparison	Compare projects by agency, borough, category, phase and reporting period.
5	Build historical understanding	Study how budgets, spending and completion dates changed over time.

6	Create a base for prediction	Use historical information later to test whether project risks can be identified in advance.
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7. Proposed Construction BI Use Cases

Use Case 1 – Project Cost Monitoring

Business question: Which projects have high budgets, high spending or significant budget changes?

Data required: Budget, spend to date, spend percentage, budget variance, agency, project and reporting period.

Planned output: Power BI can show total budget, total spending, budget utilization, remaining budget and project-level financial details.

Use Case 2 – Project Schedule Monitoring

Business question: Which projects are progressing as expected and which projects show schedule changes?

Data required: Project phase, forecast completion, actual completion where available, reporting period and schedule history.

Planned output: Power BI can show projects by phase, completion trends and historical schedule changes.

Use Case 3 – Agency and Location Analysis

Business question: How does the project portfolio differ across agencies and NYC locations?

Data required: Managing agency, sponsor agency, borough, community board, category, budget and spending.

Planned output: Users can compare project portfolios across agencies, boroughs and categories.

Use Case 4 – Historical Project Performance

Business question: How have budgets, spending and schedules changed over time?

Data required: Historical budget/spend and schedule snapshots.

Planned output: Power BI can show trends and changes across reporting periods.

8. KPI Framework

The following KPIs are proposed for the project. They are analytical measures for this internship and are not being presented as official NYC performance targets.

KPI	Simple meaning	Area
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Total Project Budget	Total budget represented by the selected project records.	Financial
Total Spend	Total spend to date in the selected records.	Financial
Budget Utilization %	Spend to date divided by total budget.	Financial
Remaining Budget	Total budget minus spend to date.	Financial
Budget Variance	Change in budget between reporting periods.	Financial
Active Projects	Count of projects currently represented as active based on available project status/phase information.	Portfolio
Projects by Phase	Count of projects across current project phases.	Schedule
Schedule Variance	Change in completion date across reporting periods where the historical data supports the calculation.	Schedule
Projects by Agency	Number and financial size of projects by managing agency.	Portfolio
Projects by Borough	Project distribution by borough.	Portfolio

9. Analytical Framework

- Document assumptions, limitations and source information so the analysis remains transparent and reproducible.
- Design a Power BI reporting structure that presents information clearly for business users.
- Define business KPIs for budget, spending, utilization, schedule and project portfolio monitoring.
- Assess data quality, completeness and consistency before using the information for reporting.
- Identify the relevant NYC capital project data and understand its structure and reporting period.
- Define the business questions and management decisions the BI solution should support.

The proposed analytical approach is:

10. Proposed Power BI Architecture

Layer	Planned use
Data Sources	NYC Open Data Capital Projects datasets.
Data Preparation	Power Query for data types, missing values, formatting, filtering and combining relevant tables.

Data Model	Project, agency, date, location, financial and schedule information, with relationships based on the actual data.
Analytics	DAX measures for project KPIs and trend analysis.
Visualization	Power BI pages for executive, financial, schedule and portfolio analysis.
Decision Support	Simple indicators and findings to help users focus on projects that need further review.

11. Data Quality Plan

- Check whether numeric fields are stored as numbers and dates are stored in a consistent format.
- Identify missing values in important project, financial and schedule fields.
- Check agency and category names for consistency.
- Check FMS IDs and understand whether repeated IDs represent different records or genuine duplicates.
- Compare selected calculations with the source data before using them in the dashboard.
- Document assumptions instead of silently changing unclear values.
- Keep a record of the dataset update date and reporting period used in the project.

12. Technology and Tool Selection

Tool / Source	Purpose
Microsoft Power BI	Main tool for data modeling, DAX, dashboards and reporting.
Power Query	Main tool for importing, cleaning and transforming the public datasets.
Excel / CSV	Possible supporting format for downloaded public data and manual checks.
NYC Open Data	Primary public data source.
GitHub	Version control and portfolio presentation for the internship work.

13. Dashboard Roadmap

Stage	Planned output
Stage 1 – Executive Overview	Projects, budget, spending, utilization and portfolio summary.

Stage 2 – Financial Analysis	Budget vs spending, utilization, variance and agency/project analysis.
Stage 3 – Schedule Analysis	Project phase, forecast completion and historical schedule movement.
Stage 4 – Portfolio Analysis	Agency, borough, project category and project-size comparisons.
Stage 5 – Risk / Predictive Analysis	Test historical patterns for possible budget or schedule risk.

14. Governance and Documentation

Since the project uses public data, the main governance focus will be transparency and reproducibility. The project will document the source, reporting period, transformations, KPI definitions, assumptions and limitations.

- Record the exact NYC Open Data dataset used.
- Record the date on which the dataset was downloaded.
- Keep the original downloaded data unchanged.
- Keep a separate cleaned version for analysis.
- Document Power Query transformation steps.
- Document DAX measures used in the dashboard.
- Clearly separate actual results from planned analysis.
- Keep project files organized so the work can be reviewed or repeated.

15. Project Limitations

The project uses public NYC data rather than private construction-company data.

The selected datasets mainly support financial and schedule analysis.

Detailed employee productivity, supplier performance, material consumption and safety information will not be invented.

Predictive analysis will only be performed if the available historical data is sufficient.

16. Conclusion

This strategy provides a clear starting point for developing a construction Business Intelligence project using real public data. The NYC Capital Projects datasets are suitable because they contain project, financial and schedule information and also provide historical records for further analysis.

The main idea is to create a clear, evidence-based BI approach in which the selected data, business questions, KPIs, data-quality checks and reporting requirements are defined before management insights are presented.

17. Official Data Sources

- NYC Open Data – Capital Projects Dashboard: Citywide Budget and Schedule
<https://data.cityofnewyork.us/City-Government/Capital-Projects-Dashboard-Citywide-Budget-and-Sch/fb86-vt7u>
- NYC Open Data – Capital Projects Dashboard: Citywide Budget and Spend by Fiscal Year
<https://data.cityofnewyork.us/City-Government/Capital-Projects-Dashboard-Citywide-Budget-and-Spe/gyhf-rsr3>
- NYC Open Data – Capital Projects Dashboard: Citywide Schedule History and Variance
<https://data.cityofnewyork.us/City-Government/Capital-Projects-Dashboard-Citywide-Schedule-Histo/95tx-snak>
- NYC Open Data – Capital Projects Dashboard: Citywide Budget Spend History and Variance
<https://data.cityofnewyork.us/City-Government/Capital-Projects-Dashboard-Citywide-Budget-Spend-H/qj5n-h5qp>

Note: Dataset details cited in this document were checked against NYC Open Data pages available in October 2026. Dataset contents and update dates may change when NYC Open Data publishes a new reporting release.